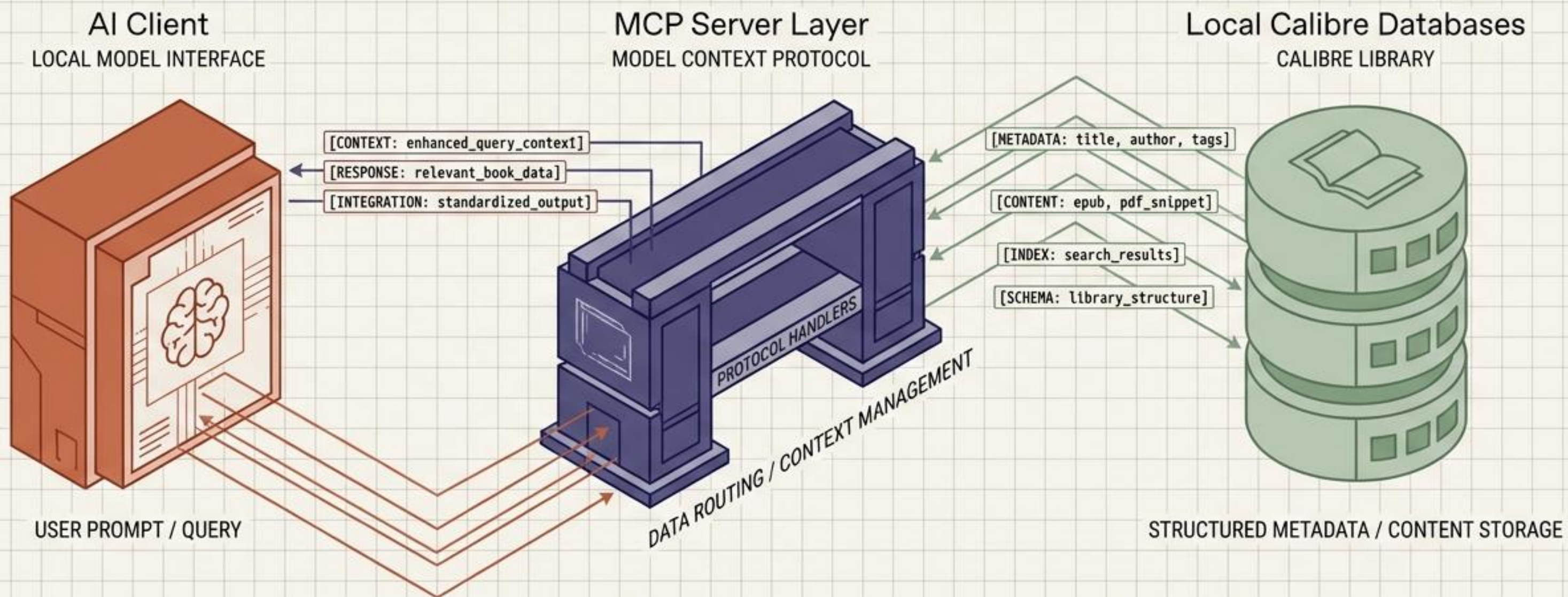


Architecting mcp-calibre

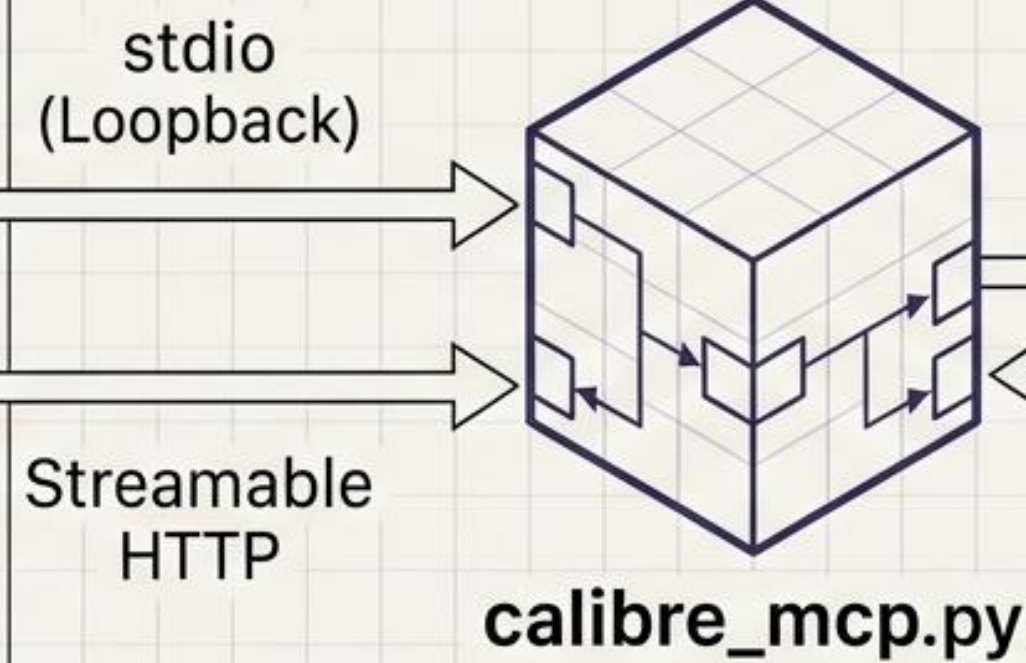
Native AI access to local libraries through the Model Context Protocol.



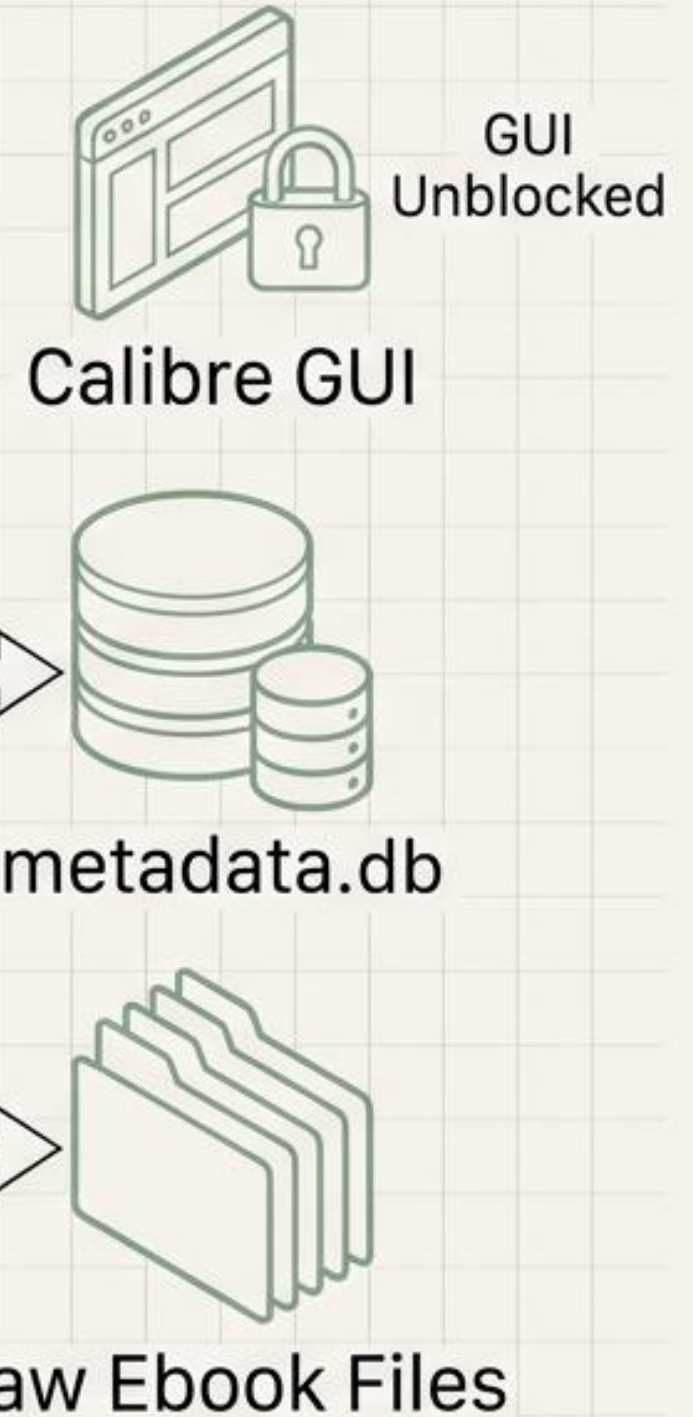
MCP Clients



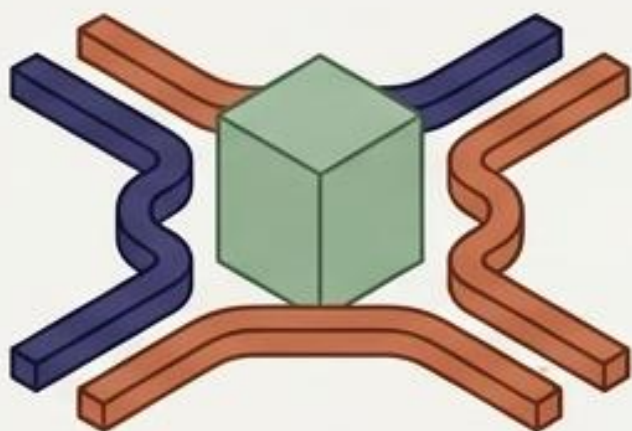
mcp-calibre Server



Local Machine



Constraint-Driven Engineering Principles



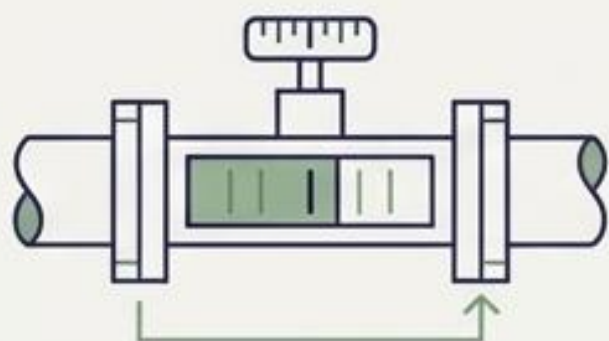
Zero Conflict

Bypasses `calibredb` subprocesses entirely. Direct SQLite queries ensure millisecond reads without locking the Calibre GUI or throwing process errors.



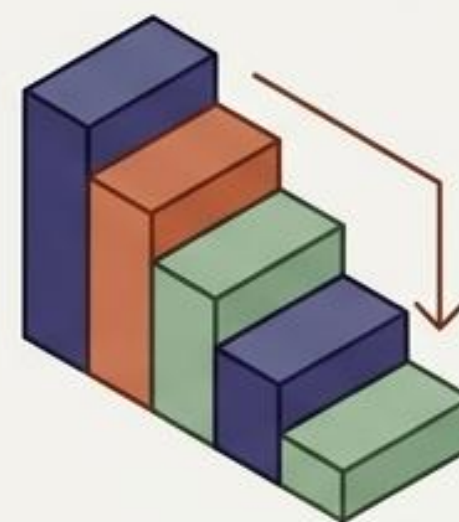
Read-Only Integrity

Enforces strict `PRAGMA query_only=1` at the SQLite connection layer. No write paths to the library exist; sidecar files are quarantined in `%LOCALAPPDATA%`.



Token Efficiency

Hard caps text extraction (default 12,000 characters per read) and strictly offloads image data to local UI frames, shielding the LLM context window.



Graceful Degradation

Fails safely across built-in native Python parsers to slow subprocess fallbacks (ebook-convert), prioritizing speed but guaranteeing eventual retrieval.

Unlocking the Vault with a Sidecar Index

The Problem



Calibre Native fts5 Table

Stock SQLite cannot query the native full-text search because it utilizes a custom C-extension tokenizer.

The Solution

Read plain-text
books_text table



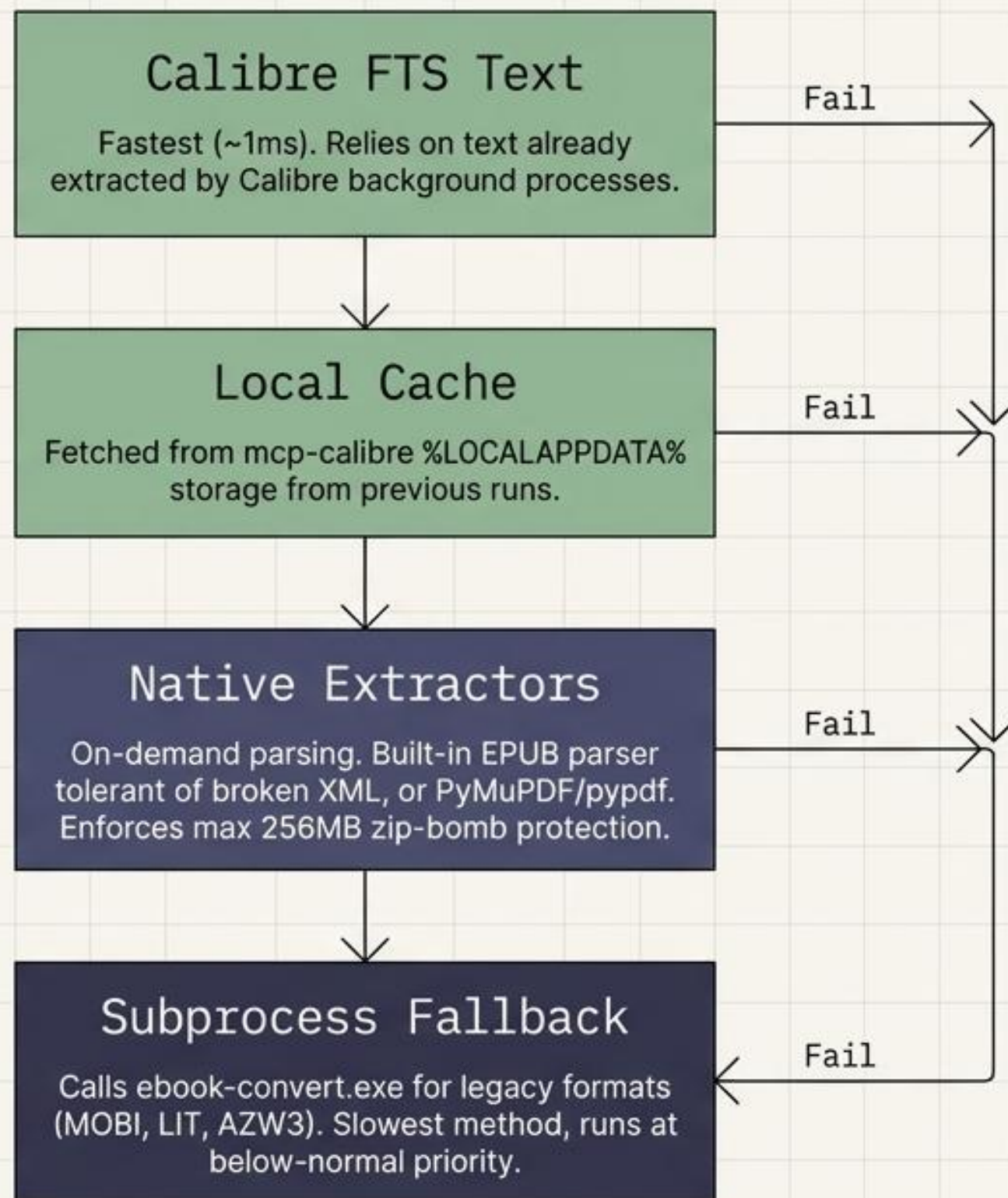
Build index.db
Sidecar

tokenizer='unicode61
remove_diacritics 2'

Incremental Sync: Background thread syncs via `text\_hash` every 10 minutes.

Storage Optimization: Utilizes `contentless\_delete=1` on SQLite ≥ 3.43 to drastically reduce storage footprint without duplicating book text.

The Text Extraction Waterfall

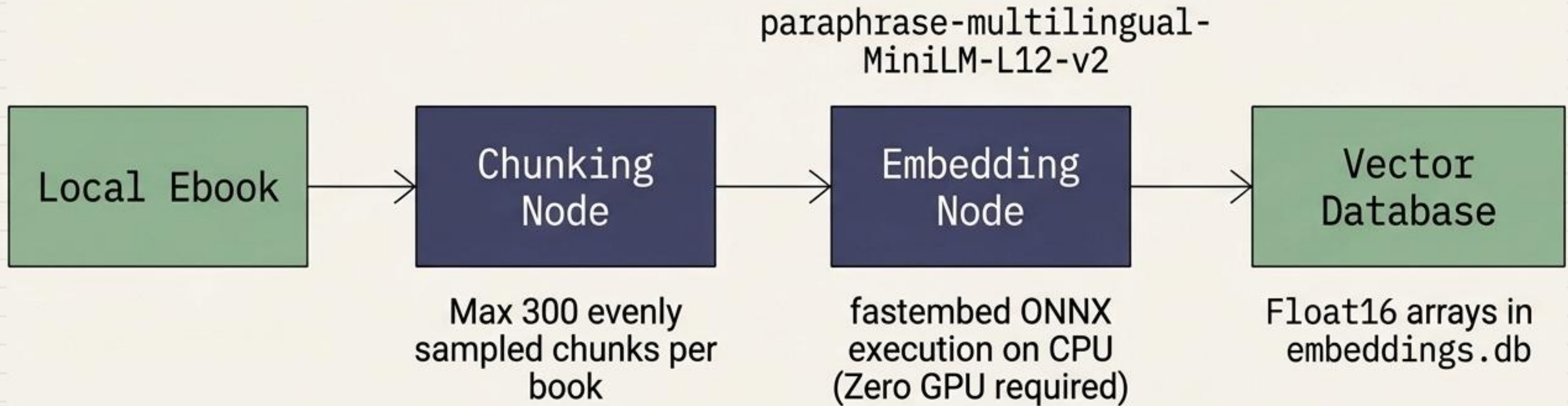


Tri-Engine Search Architecture

	Metadata Search	Lexical FTS	Semantic Search
Backend Engine	SQLite structured queries tag:security	fts5 BM25 Sidecar	fastembed ONNX vectors
Matching Logic	Exact match & Regex (tag:security)	Keyword matching (all, any, phrase, raw)	Meaning-based concept matching
Language Support	Exact string	Strictly Language- Dependent	Multilingual (e.g., IT queries find EN text)
Latency	~5ms	80-90ms	Computationally heavy, but local

Synthesis Insight: Layering traditional BM25 keyword matching with vector-based semantic search guarantees retrieval regardless of the LLM's query phrasing or language.

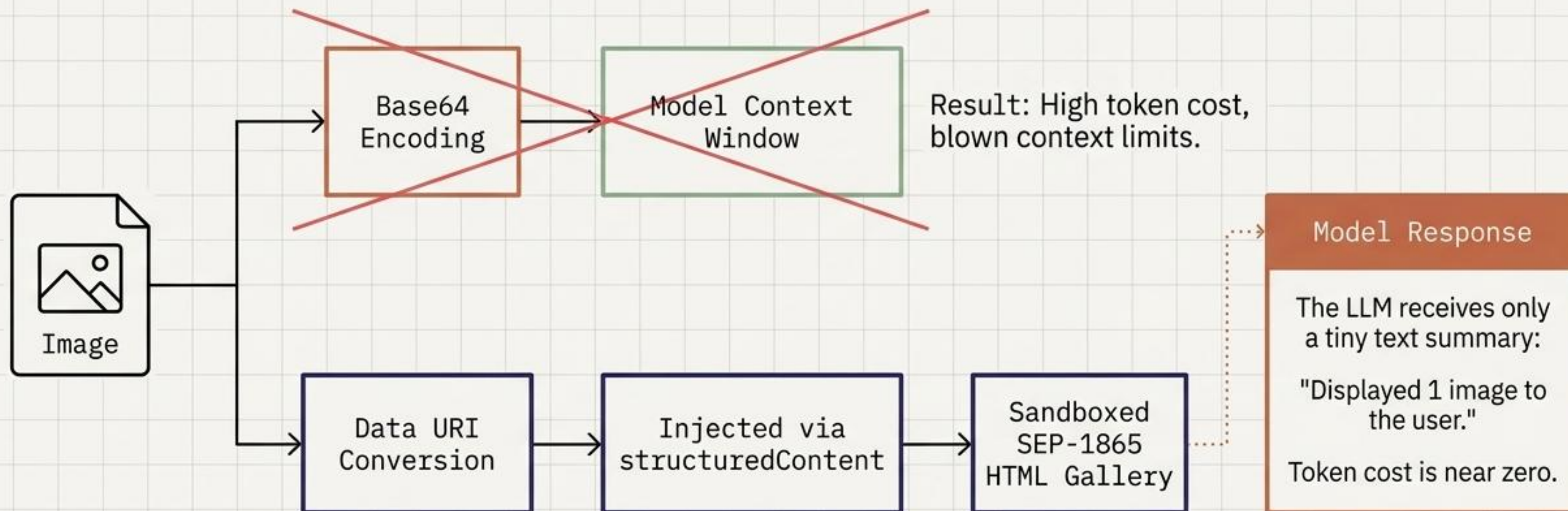
Air-Gapped Semantic Search Subsystem



Security Posture: Weights are downloaded once from Hugging Face during setup. Queries and book data never leave the local machine. Zero external API calls.

Shielding the Context Window with MCP Apps

Scenario: The LLM is asked to "Show Figure 3"



Note: Enforces a restrictive Content Security Policy (CSP) blocking network access. Native clipboard API fallback preserves Copy PNG functionality.

Transport Boundaries & Attack Surface Mitigation

stdio Transport

Use Case:

Local desktop agents (Codex, Claude Desktop)

Security Model: Tightly coupled architecture relying on implicit local machine trust.

Streamable HTTP Transport

Use Case:

LAN sharing, remote execution, Claude Code

Security Model: Stateless JSON responses.

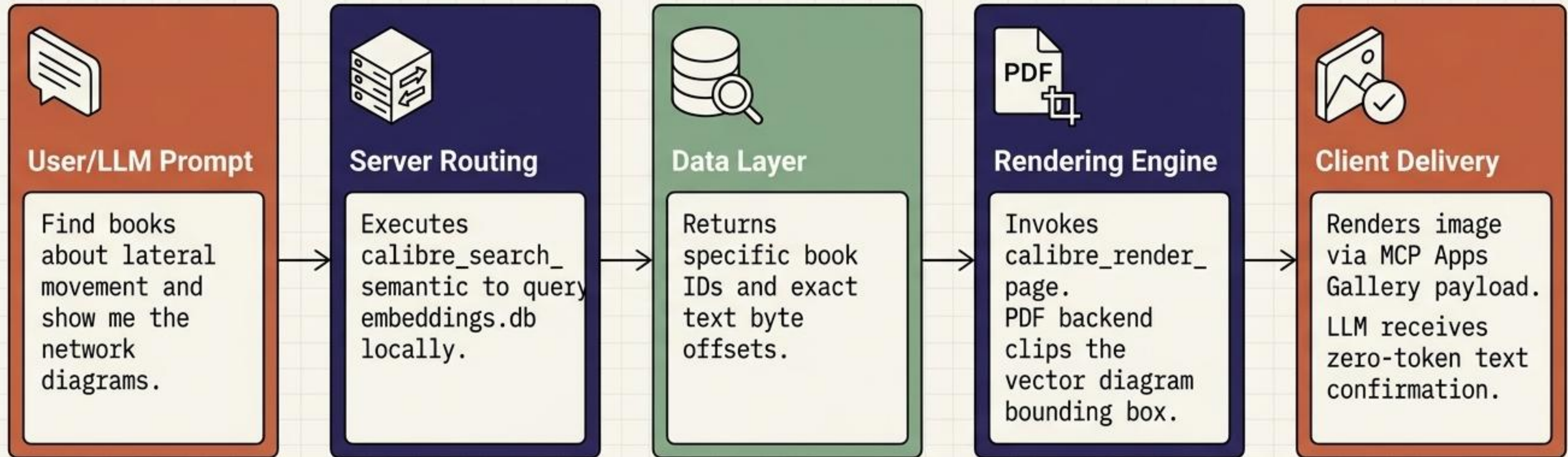
Authentication: Mandatory Bearer auth (minimum 24 chars) utilizing constant-time comparison (`hmac.compare_digest`).

Network: DNS rebinding protection via strict Host-header allow-lists. Default loopback binding.

Shared Sandbox Boundary: Path Confinement

File paths derived from the database are actively resolved and aggressively rejected if they attempt to traverse outside the designated Calibre library root directory.

Architectural Synthesis: The Execution Flow



Takeaway: By decoupling text extraction, vector embeddings, and UI payloads from the core LLM context, legacy offline files are seamlessly synthesized into a modern AI-native experience without latency or token bloat.